

Notes on grifting by optimal transport

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Consider an affine measurable space $\mathcal{S} \subset \mathbb{R}^d$. Fixing $(\Omega, \Sigma, \mathbb{P})$, $\Omega = \mathcal{S}$, $\Sigma = \mathcal{B}(\mathcal{S})$ the Borell σ -algebra of $\mathcal{S}$, and $\mathbb{P}$ a probability measure on (Ω, Σ) . Let $v := (v_t)_{t \in \mathbb{N}}$ be a collection of Σ -measurable automorphisms on $\mathcal{S}$.

The dynamical system of a drifting model (Deng et al., 2026) evolves the prediction during training according to the following Markov chain:

$$X_{t+1}^v = X_t^v + v_t(X_t) \in \mathcal{S} \quad (1)$$

equivalently, letting μ_0 be probability measure on (Ω, Σ) , this can be viewed in terms of the evolution of the law of X_t from its initial distribution μ_0 as

$$\begin{cases} \mathcal{L}(X_{t+1}^v) = (\text{Id} + v_t)_\# \mathcal{L}(X_t^v) \\ \mathcal{L}(X_0^v) = \mu_0 \end{cases} \quad (2)$$

The development of drifting models Deng et al. (2026) rests on the idea of crafting (by hand) a vector-field-valued map v which is asymptotically stable and whose attracting equilibrium induces $X_\infty \sim \nu$ for a given target distribution $\nu \in \mathcal{P}(\mathcal{S})$. By using the methodology of least-squares fitting outlined in Deng et al. (2026), the specification of v can be leveraged into the *training* of a neural network f which yields an approximate pushforward map transforming a given noise distribution μ_0 into the target distribution ν , i.e. $\nu \approx f_\# \mu_0$.

From a technical standpoint, the authors of Deng et al. (2026) construct a family of vector fields v depending on ν and $\mathcal{L}(X_t)$ which admits as an equilibrium the target distribution ν . However, under general conditions it remains unclear whether the resulting mean-field dynamics of X_t will meaningfully converge to ν as $t \rightarrow \infty$, despite them giving some sufficient conditions.

A contrario, it is tempting to reinterpret (1) as a control problem, and try and determine v through the specification of an objective function to be optimised. At an intuitive level, we can then view the efforts of Deng et al. (2026) as an ad-hoc attempt to guess a control v which will yield the desired target distribution ν .

Inspired by the recent work of Müller et al. (2026), we propose to study this as a *measure constrained* optimal control problem.

1. Proposed setting

Let $\gamma \in [0, 1]$, $c : \mathcal{S}^2 \rightarrow \mathbb{R}$ be a measurable cost function, and let $\mu_0 \in \mathcal{P}(\mathcal{S})$ be a given initial distribution, and consider the optimal control problem,

$$\begin{aligned} J_p &:= \min_{v \in \mathcal{V}} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t c(X_t^v, X_{t+1}^v) \right] \\ &\text{subject to } X_{t+1}^v = X_t^v + v_t(X_t) \text{ for all } t \geq 0, \end{aligned} \quad (3)$$

$$X_0 \sim \mu_0,$$

where $\mathcal{V} \subset (\mathcal{S}^{\mathcal{S}})^{\mathbb{N}}$ is a set of admissible controls. This problem is highly non-linear and non-smooth in v in general, which complicates resolution. While the primal theory which tackles (3) head-on, we will instead focus on the (control-theoretic) dual formulation of the problem, which is linear in the space of measures.

To do so, let $\mathcal{M}_+(\mathcal{S}^2)$ be the space of σ -finite positive measures on $\mathcal{S}^2$, and let $\mathcal{P}(\mathcal{S}^2) \subset \mathcal{M}(\mathcal{S}^2)$ be the space of probability measures on $\mathcal{S}^2$. We define the “forward” and “backward” marginalisation operators $\mathcal{F}$ and $\mathcal{B}$ on $\mathcal{M}_+(\mathcal{S}^2)$ as follows:

$$\mathcal{F} : \mu \in \mathcal{M}_+(\mathcal{S}^2) \rightarrow \int_{\mathcal{S}} \mu(dx, \cdot) \in \mathcal{M}_+(\mathcal{S}) \quad (4)$$

$$\mathcal{B} : \mu \in \mathcal{M}_+(\mathcal{S}^2) \rightarrow \int_{\mathcal{S}} \mu(\cdot, dy) \in \mathcal{M}_+(\mathcal{S}) \quad (5)$$

For any $\gamma \in [0, 1)$, let $\mathcal{P}_\gamma(\mathcal{S}^2) := (1 - \gamma)^{-1} \mathcal{P}(\mathcal{S}^2)$ be the subspace of $\mathcal{M}_+(\mathcal{S}^2)$ of measures with total mass $(1 - \gamma)^{-1}$.

For any $v \in \mathcal{V}$, with μ_0 implicitly fixed, let $\mu^v \in \mathcal{P}_\gamma(\mathcal{S}^2)$ be the measure defined by

$$\mu^v(dx, dy) := \sum_{t=0}^{\infty} \gamma^t \mathcal{L}(X_t^v, X_{t+1}^v)(dx, dy). \quad (6)$$

This is equivalently written as

$$\mu^v : (A, B) \in \Sigma^2 \rightarrow \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(X_t^v \in A, X_{t+1}^v \in B) = \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(X_t^v \in A, (\text{Id} + v_t)(X_t^v) \in B). \quad (7)$$

The *control-theoretic dual* is the linear problem

$$J_d := \min_{v \in \mathcal{V}} \int c(x, y) \mu^v(dx, dy) \quad (8)$$

$$\begin{aligned} \text{subject to } \mu^v &\in \mathcal{P}_\gamma(\mathcal{S}^2), \\ \mathcal{B}\mu^v &= \mu_0 + \gamma \mathcal{F}\mu^v \end{aligned} \quad (9)$$

$$\mathcal{F}\mu^v = \sum_{t=0}^{\infty} \gamma^t \bigcirc_{s=1}^t (\text{Id} + v_s) |_{\#} \mu_0$$

Lemma 1 *The control-theoretic duality is strong, i.e., $J_p = J_d$. (We will treat minimiser equivalence later.)*

Proof That $\int c d\mu^v = \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t c(X_t^v, X_{t+1}^v)]$ is immediate from the definition of μ^v .

To establish the constraints of the dual, note that, by definition $\mathcal{B}\mu^v = \sum_{t=0}^{\infty} \gamma^t \mathcal{L}(X_t^v)$, and thus

$$\mathcal{F}\mu^v = \sum_{t=0}^{\infty} \gamma^t \mathcal{L}(X_{t+1}^v) = \frac{1}{\gamma} (\mathcal{B}\mu^v - \mu_0). \quad (10)$$

This establishes the constraint associated to $X_0 \sim \mu_0$. The constraint for the dynamics directly leads to

$$\mathcal{F}\mu^v = \sum_{t=0}^{\infty} \gamma^t \mathcal{L}(X_{t+1}^v) = \sum_{t=0}^{\infty} \gamma^t (\text{Id} + v_t) |_{\#} \mathcal{L}(X_t^v) = \sum_{t=0}^{\infty} \gamma^t \bigcirc_{s=1}^t (\text{Id} + v_s) |_{\#} \mu_0. \quad (11)$$

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References

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